Article

Emergent Lorentz Symmetry and the Gravity Channel in a Deterministic Brane-Lattice Substrate (Paper II)

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Abstract

We develop two consistency bridges for the deterministic 4D brane-lattice substrate of Paper I. The Lorentz bridge shows that inside observers built from the same substrate material would experience effective Lorentz kinematics despite the cubic lattice being anisotropic in the lab frame. The key mechanism is a dual-observer argument: both signal cones and rod-shaped measuring devices renormalize with the same branch-speed anisotropy, so the ratio distance/transit-time is direction-independent to leading order. Residual birefringence—the splitting between effective transverse and longitudinal signal speeds that survives at finite wavelength—is the primary falsifiable prediction and can be computed directly from the stiffness tensor. The gravity bridge identifies the same geometric mechanism, the induced-metric term $\partial_i u\, \partial_j u$, as the substrate origin of a long-range attractive channel: localized transverse excitation sources a slowly varying in-brane contraction field that biases nearby wave-packet propagation inward. In a full 4D treatment, spacelike excitation lengthens the timelike link (time dilation) and timelike excitation lengthens spacelike links (length contraction / gravity channel)—two faces of one Pythagorean rule. The open quantitative task is to derive a Newtonian-limit Poisson equation from the coarse-grained contraction field and express the effective gravitational coupling G_{eff} in substrate parameters (k_s, a, α, m).

Keywords: deterministic substrate; brane lattice; emergent Lorentz symmetry; dual-observer argument; birefringence; gravity channel; transverse-lateral coupling; induced metric; Newtonian limit

1. Introduction

This is the second paper in a series developing a deterministic brane-lattice substrate hypothesis. Paper I established the substrate model — a 4D cubic lattice embedded with codimension zero in a 4D Euclidean ambient, governed by a hyperelastic action built from the induced metric — and derived the linearized wave-branch structure with characteristic speeds $c_T^2 = (1-\alpha)k_s a^2/m$ and $c_L^2 = k_s a^2/m$. The present paper develops two consistency bridges from that foundation.

What this paper imports. Both bridges import from Paper I the following interface equations: the near-identity metric expansion (Equation (29)), the linearized branch speeds (Equation (17)), the long-wavelength dispersion (Equation (1)), and the quasi-static in-brane equilibrium equations (Section 3.9). No new substrate assumptions are introduced here.

Lorentz bridge. The cubic lattice is manifestly anisotropic at the lab-frame level: the longitudinal wave speed along [100] differs from the speed along [111] by roughly 7.5% at $\alpha = 0.2$. The Lorentz bridge (Section 3) argues that this anisotropy cancels for inside

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observers whose rods, clocks, and signal cones are made of the same substrate material. The cancellation is not tuned; it is a structural consequence of the substrate-universality principle: the same stiffness tensor that sets branch speed also determines the elastic response of any rod-shaped excitation. Residual birefringence — the splitting of effective signal speeds that survives at finite wavelength and amplitude — is the primary falsifiable prediction.

Gravity bridge. The induced metric $g_{ij} = h_\star^2\delta_{ij} + \partial_i u\,\partial_j u$ contains a quadratic source term in the transverse gradient. When the in-brane displacement field $\vec{\xi}$ is free to respond, quasi-static balance forces a slowly varying contraction pattern inward toward any region of localized transverse excitation. The gravity bridge (Section 4) identifies this as the substrate origin of a long-range attractive channel. The bridge is geometrically fixed; the open quantitative task is to derive a Newtonian-limit Poisson equation from the coarse-grained contraction field and express the effective coupling G_{eff} in substrate parameters.

Why these two bridges are paired. Both bridges operate through the same geometric mechanism — the Pythagorean lengthening of links by transverse displacement — and in the fully symmetric 4D treatment they are two faces of one rule: spacelike excitation lengthens timelike links (time dilation) and timelike excitation lengthens spacelike links (gravity channel / length contraction). Pairing them in one paper makes this unification explicit and avoids treating the two phenomena as independent postulates.

Paper organization. Section 2 lists the interface equations imported from Paper I. Section 3 develops the Lorentz bridge: wave cone, dual-observer argument, analogue-metric viewpoint, birefringence as falsifiable signature, and open problems. Section 4 develops the gravity bridge: contraction-channel derivation, Newtonian-limit open problem, and simulation test specification.

2. Imported interfaces from Paper I

This paper imports the following results from Paper I without re-derivation. All equation labels refer to Paper I unless otherwise indicated.

Substrate model. The 4D cubic brane lattice, its embedding map $\mathbf{X} : \Omega \times \mathbb{R} \to \mathbb{R}^4$, the discrete lattice energy (Equation (7)), and its continuum limit (Equation (24)).

Induced metric. $g_{ij} = \partial_i \mathbf{X} \cdot \partial_j \mathbf{X}$ (Equation (10)), and its near-identity expansion around a homogeneous background (Equation (29)):

$$g_{ij} = h_\star^2\delta_{ij} + h_\star(\partial_i\xi_j + \partial_j\xi_i) + \partial_i\vec{\xi}\cdot\partial_j\vec{\xi} + \partial_i u\,\partial_j u. \tag{P1:29}$$

Equations of motion. Equation (26) of Paper I.

Linearized branch speeds. $c_T^2 = (1-\alpha)\,k_s a^2/m$ and $c_L^2 = k_s a^2/m$ from Equation (17).

Long-wavelength dispersion. $\omega_b^2(\mathbf{k}) \approx c_b^2\,|\mathbf{k}|^2$ for $|\mathbf{k}| \to 0$ (Equation (1)).

Quasi-static in-brane equilibrium and contraction channel. Section 3.9 of Paper I, which derives the in-brane forcing problem sourced by $\partial_i u\,\partial_j u$.

Bell interpretive stance. The retrocausal worldtube reading from Section 5 of Paper I, imported as a fixed interpretive commitment.

3. Lorentz bridge: effective kinematics for inside observers

This bridge imports the linearized branch speeds $c_L^2 = k_s a^2/m$ and $c_T^2 = (1-\alpha)\,k_s a^2/m$ from Paper I (Equation (17)) and asks whether inside observers built from the same substrate material would experience these as a universal, direction-independent

Lorentz invariant kinematics. The ambient lattice is cubic and is *not* rotationally symmetric; the bridge argument must therefore close the gap between lab-frame anisotropy and observer-frame isotropy.

3.1. Linear-wave cone and effective Lorentz kinematics

The substrate carries Lorentzian sign structure only at the level of the brane action; the ambient $\mathbb{R}^4$ in which the brane is embedded is symmetric Euclidean. A Lorentz metric on excitations is therefore not postulated — it is an emergent property of the linear wave regime as seen by an inside observer.

From Paper I, linearizing the equations of motion about a homogeneous background yields the long-wavelength dispersion

$$\omega_b^2(\mathbf{k}) \;\approx\; c_b^2\,|\mathbf{k}|^2 \quad \text{for } |\mathbf{k}| \to 0, \tag{1}$$

which defines an effective wave cone for each polarization branch b. In regimes where a single branch dominates transport and dispersion is negligible, the wave equation is approximately invariant under the standard Lorentz group at speed $c \equiv c_T$, with boost parameter

$$\gamma = \left(1 - v^2/c^2\right)^{-1/2}, \tag{2}$$

as an effective symmetry of the linear propagation sector, not as a microscopic postulate.

3.2. Dual-observer argument: lab anisotropy vs internal isotropy

On the 6-neighbor axial-only lattice the long-wavelength wave speed is direction-dependent at the lab-frame level — for example $c_L([100])/c_L([111]) \approx 1.075$ at the default operating point $\alpha = 0.2$. The relativistic effective description (2) is therefore *not* a statement about the lab frame; it is a conjecture about *internal* observers, whose rods, clocks, and signal cones are substrate-emergent.

The closing mechanism is as follows. A rod built from substrate material is itself a strain pattern propagating at the same branch speeds as the signal. An inside observer's unit-length standard rod in direction $\hat{n}$ therefore has a physical length (in the lab metric) that scales with $c(\hat{n})$ — exactly the same direction dependence as the signal cone in that direction. When both the signal and the measuring rod renormalize with the same anisotropy factor, the observed ratio (distance/signal-transit-time) is direction-independent to leading order.

This is not a tunable coincidence. It is a structural consequence of the principle that observers and signals are made of the same substrate material: the stiffness tensor that sets the branch speed also sets the elastic response of a rod-shaped excitation, so anisotropy factors cancel in any local observer measurement.

Falsifiable residue. The cancellation is exact only in the strict long-wavelength, single-branch, weak-amplitude limit. At finite wavelength and nonzero mixing between branches, a residual direction dependence survives. Its magnitude at scale ka is a calculable quantity that depends on the stiffness tensor and neighbor stencil. *Residual birefringence* — a direction-dependent splitting of effective light speed — is therefore the primary falsifiable signature of the bridge: if two solitons in different orientations measure the same effective signal speed to better than the residual predicted by the stiffness tensor, the bridge is supported; if they do not, it fails.

3.3. Analogue-metric viewpoint

An equivalent viewpoint, standard in analogue-gravity systems, is that linear excitations experience an effective spacetime interval

$$ds^2 = -c^2\, dt^2 + g_{ij}^{(\mathrm{eff})}(x,t)\, dx^i\, dx^j, \tag{3}$$

where $g_{ij}^{(\mathrm{eff})}$ depends on the background configuration about which one linearizes. Two geometric objects contribute to this background: the induced brane metric $g_{ij}(\mathbf{X})$ from Paper I (10), and the branch-dependent effective coefficients of the linearized wave operator. Slowly varying transverse profiles $u = X^4$ modify intrinsic distances at the level of the induced metric through the term $\partial_i u\, \partial_j u$, and can therefore be read as a background that focuses or defocuses wave propagation in close analogy to a weak gravitational field. Both encode how background deformations bend wave transport, as in analogue-gravity constructions [1,2].

3.4. Regime of validity and expected deviations

Effective Lorentz symmetry is expected to break when: (i) multiple branches contribute comparably (mode mixing); (ii) dispersion becomes significant at shorter wavelengths; (iii) geometric or material nonlinearity becomes important at larger amplitudes.

In the present theory these deviations are structural consequences of a substrate whose timelike direction is selected by the brane action rather than imposed as a metric primitive. They delimit the domain where a relativistic effective description applies, and each regime transition is a calculable correction to the dual-observer argument, not an unfalsifiable escape hatch.

3.5. Lorentz priority open problems

1. **Observer-universality calculation.** Compute the direction-dependent wave speed on the 6-neighbor cubic stencil at finite ka and determine the magnitude of residual birefringence as a function of wavelength and α. This bounds from above the scale at which effective Lorentz invariance is expected.
2. **Two-soliton isotropy test.** Run two solitons in orthogonal directions and compare their effective internal metric (signal-transit-time ratios). Agreement to within the predicted residual birefringence is the decisive numerical check.
3. **Multi-stencil tuning.** Explore whether shell weights on the cubic stencil can be adjusted to suppress residual anisotropy further, and at what cost to other bridge properties (e.g. wave speed splitting).

4. Gravity bridge: contraction channel from transverse excitation

This bridge imports the near-identity metric expansion (29) and the quasi-static in-brane equilibrium from Paper I (Section 3.9) and asks whether the resulting slowly varying contraction field can serve as the quantitative substrate of a Newtonian gravitational potential. The geometric mechanism is fixed by Paper I; the open problem addressed here is its quantitative reduction.

4.1. Recap: transverse excitation sources lateral contraction

In the near-identity decomposition around a homogeneous background $\mathbf{X}_\star(x) = (h_\star x, 0)$, the induced metric expansion is (Equation (29) of Paper I):

$$g_{ij} = h_\star^2 \delta_{ij} + h_\star(\partial_i \xi_j + \partial_j \xi_i) + \partial_i \vec{\xi} \cdot \partial_j \vec{\xi} + \partial_i u\, \partial_j u. \tag{4}$$

Setting $\vec{\xi} \equiv 0$ gives the graph picture; the gravity channel requires $\vec{\xi} \neq 0$, i.e. the in-brane displacement must be free to respond to the source term.

In the small-slope regime the in-brane strain is

$$\varepsilon_{ij} \approx \tfrac{1}{2}(\partial_i \xi_j + \partial_j \xi_i) + \tfrac{1}{2}\,\partial_i u\,\partial_j u, \tag{5}$$

so quasi-static balance $\partial_j \sigma_{ij} = 0$ with $\sigma_{ij} = \lambda\,\mathrm{tr}(\varepsilon)\delta_{ij} + 2\mu\,\varepsilon_{ij}$ gives a forcing/eigenstrain problem for the slow field $\vec{\xi}$:

$$\partial_j\big[\lambda\,(\partial_k \xi_k)\delta_{ij} + \mu(\partial_j \xi_i + \partial_i \xi_j)\big] = -\tfrac{1}{2}\partial_j\big[\lambda\,|\nabla u|^2 \delta_{ij} + \mu(\partial_i u \partial_j u + \partial_j u \partial_i u)\big]. \tag{6}$$

The right-hand side is the effective body force sourced by a transverse profile $u(x)$. The solution $\vec{\xi}$ is the in-brane contraction field; its divergence $\partial_i \xi_i$ is the local volume contraction (dilation) pattern.

Sign: always inward under pre-tension. For an isotropic pre-tension $\alpha < 1$, additional transverse arc length increases elastic energy and therefore tensile stress, which produces net lateral forces pulling material points *toward* the region of transverse excitation. For a localized bump or rotating transverse pattern, the surrounding brane develops an inward bias (contraction) rather than an outward pressure. The sign is fixed by geometry and pre-tension and does not require a separate assumption.

4.2. Unification: length contraction and time dilation

The Pythagorean argument is direction-blind. In the underlying 4D ontology (Paper I Section 3.1), any displacement perpendicular to a link stretches that link, regardless of which ambient direction is perpendicular. The same mechanism, applied symmetrically across all four directions, predicts two consequences:

1. **Spatial contraction (gravity channel):** Displacements in the timelike direction ($X^4 = u$) lengthen spacelike links — which is what the present section quantifies.
2. **Time dilation:** Displacements in the three spacelike directions (in-brane kinetic energy / spatial motion) lengthen the link along the timelike direction. Length contraction and time dilation are therefore two faces of one rule; a full 4D treatment of the brane action would express both in a single expression.

4.3. Contraction field as a gravity-like background

Linear wave packets propagating on top of the slowly varying $\vec{\xi}$-background experience branch-dependent effective coefficients in the linearized wave operator. In the effective-metric viewpoint (Bridge 1, Section 3), the $\vec{\xi}$-sourced dilation pattern acts as an analogue curved spacetime that focuses or defocuses wave transport.

Concretely, a transverse excitation concentrated in a region $\mathcal{V}$ produces through (6) an outward-decaying contraction field $\partial_i \xi_i(x)$ that attracts nearby wave packets inward. In the weak-field, slowly varying limit, this biases propagation in precisely the way a Newtonian potential would.

4.4. Open quantitative reduction: Newtonian limit

What remains is the quantitative reduction. The goal is to derive a scalar potential $\Phi(x)$ from the coarse-grained in-brane strain or elastic energy density such that

$$\nabla^2 \Phi = 4\pi G_{\mathrm{eff}}\,\rho_{\mathrm{excitation}}, \tag{7}$$

with a coupling G_{eff} expressed in substrate parameters (k_s, a, α, m). The specific steps required:

1. **Solve the forcing problem (6) in a scalar approximation.** For a spherically symmetric transverse profile $u(r)$, symmetry reduces (6) to a scalar ODE for $\partial_i \xi_i$.
2. **Identify the effective potential.** Show that $\Phi \propto \partial_i \xi_i$ (or a suitable Green's function convolution thereof) satisfies a Poisson-type equation with a source proportional to the transverse energy density $\frac{1}{2}|\nabla u|^2$.
3. **Match to Newtonian gravity.** Express G_{eff} in terms of k_s, a, α, m; derive the dimensional prediction $G_{\text{eff}} \sim k_s a^5 / m^2$ (or similar) that sets the scale of the coupling.
4. **Numerical check.** Simulate a localized transverse excitation, measure the induced $\vec{\xi}$-field, and compare its spatial decay to the Green's function prediction. Inject a long-wavelength probe packet and measure whether its trajectory is deflected inward at the rate predicted by Φ.

4.5. Gravity bridge priority open problems

1. **Scalar reduction of (6).** Derive the spherically symmetric Poisson equation for the contraction field and obtain G_{eff} in closed form.
2. **Contraction-channel extraction (numerical).** Create a localized transverse Gaussian excitation; measure the induced slow $\partial_i \xi_i$-field; compare its spatial profile to the prediction from step 1.
3. **Probe-packet deflection test.** Inject a long-wavelength wave packet, measure the lateral deflection, and check that the deflection angle matches the effective potential derived in step 1.
4. **4D symmetric treatment.** Extend the slice-restricted analysis to a 4D treatment of the brane action, expressing both length contraction and time dilation in a single expression, and derive the full stress-energy coupling tensor.

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Appendix A. Symbol dictionary

$\Omega \subset \mathbb{R}^3$	Intrinsic/material domain of the brane (coordinates $x = (x^1, x^2, x^3)$).
$\mathbf{X}(x,t) \in \mathbb{R}^4$	Embedding field of the brane into ambient 4D Euclidean space.
t	Slicing parameter along the inside observer's timelike direction ($t := x^4$); not an external/ambient time (the ambient is symmetric 4D Euclidean, Section 3.1).
g_{ij}	Induced metric: $g_{ij} = \partial_i \mathbf{X} \cdot \partial_j \mathbf{X}$.
g^0_{ij}	Isotropic reference metric: $g^0_{ij} = \ell_0^2\, \delta_{ij}$ (stress-free scale ℓ_0).
$h_\star$	Held ground-state scale (e.g. imposed by boundary conditions).
α	Dimensionless pre-stress parameter: $\alpha = \ell_0/h_\star \in (0,1]$.
$\widehat{g}$	Mixed relative metric: $\widehat{g} = (g^0)^{-1} g$.
E	Green–Lagrange strain: $E = \frac{1}{2}(\widehat{g} - I)$.
λ, μ	Effective Lamé parameters derived from the spring constitutive law: $\mu(\alpha) = (1-\alpha)k_s/a$, $\lambda = 0$ on the 6-neighbor stencil. StVK agrees at quadratic order only; see Section 3.5.
ρ_m	Brane mass density in the Lagrangian.
$W(g; g^0)$	Isotropic stored-energy density (hyperelastic).
$P_A^{\ i}$	First Piola stress: $P_A^{\ i} = \partial W / \partial(\partial_i X^A)$.
$\mathbf{u}$	Displacement field relative to a background: $\mathbf{u} = \mathbf{X} - \mathbf{X}_\star$.
u	Shorthand for a transverse component in a graph-type embedding, $\mathbf{X} \approx (x, u)$.
$\boldsymbol{\epsilon}$	Normalized polarization state of a narrowband multi-component wave.
a_μ	Berry (rank-1) connection: $a_\mu = \mathrm{i}\langle u \,\vert\, \partial_\mu u \rangle$.
$f_{\mu\nu}$	Berry curvature: $f_{\mu\nu} = \partial_\mu a_\nu - \partial_\nu a_\mu$.
$\mathcal{A}_\mu$	Wilczek–Zee connection on an n-dimensional polarization subspace.
$\mathcal{F}_{\mu\nu}$	Wilczek–Zee curvature: $\partial_\mu \mathcal{A}_\nu - \partial_\nu \mathcal{A}_\mu - \mathrm{i}[\mathcal{A}_\mu, \mathcal{A}_\nu]$.
γ_Γ	Geometric phase / holonomy around loop Γ: $\gamma_\Gamma = \oint_\Gamma a_\mu\, \mathrm{d}x^\mu$.
Ω_{mix}	Mode-mixing (non-adiabatic conversion) rate between nearby polarization bands.
$\Delta\omega_{\mathrm{gap}}$	Local spectral gap to the nearest competing polarization band.
a	Lattice spacing of the cubic substrate (microphysical length scale).
$\tilde{a}$	Momentum-lattice spacing in the discretized Brillouin zone (typically $\tilde{a} \sim 2\pi/(N_s a)$).
$\mathbf{k}$	Crystal momentum / Brillouin-zone coordinate for lattice eigenmodes.
$\Psi = (\psi_x, \psi_y, \psi_z)^\top$	Axis-aligned complex narrowband triplet carrier (cubic lattice internal sector).
T^a	Generators of $\mathfrak{su}(3)$ (e.g. Gell–Mann basis) used to decompose a $U(3)$ connection.
$R_p^l \in \mathbb{R}^4$	4D node position in the explicit lattice: $p = (i,j,k)$ spacelike triple, $l \in \mathbb{Z}$ timelike index.
k_s	Central-force spring stiffness of the 6-neighbor spacelike stencil.
m	Node mass (discrete lattice; continuum mass density $\rho_m = m/a^3$).
Δt	Temporal lattice step (timelike link length in the explicit 4D lattice).
$\mathcal{N}_s$	6-neighbor axial spacelike stencil: $\{\pm\hat{e}_x, \pm\hat{e}_y, \pm\hat{e}_z\}$.
$\mathcal{N}_t$	Temporal neighbor stencil: $\{l \pm 1\}$.
c_L	Long-wavelength longitudinal wave speed: $c_L^2 = k_s a^2/m$.
c_T	Long-wavelength transverse wave speed: $c_T^2 = (1-\alpha)\, k_s a^2/m$.

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